Abstract:

Breast cancer is one of the most prevalent and life-threatening diseases affecting women worldwide. Early detection and accurate diagnosis play a crucial role in improving patient survival rates and enabling timely treatment decisions. This project focuses on developing and evaluating machine learning models for the classification of breast cancer tumors as benign or malignant using the Breast Cancer Wisconsin (Diagnostic) dataset from the UCI Machine Learning Repository. The dataset consists of 569 samples and 30 features computed from digitized images of fine-needle aspirate (FNA) tests of breast masses.

In this study, four supervised learning algorithms—Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Random Forest—were implemented and compared based on their classification performance. Data preprocessing involves feature standardization to ensure consistency and improve model convergence. The dataset was divided into training and testing subsets using an 80:20 split. Model evaluation was conducted using performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC values.

The experimental results revealed that Logistic Regression achieved the highest accuracy of 98.25% and an ROC-AUC of 0.995, followed closely by SVM with similar results. Random Forest and KNN also performed well, achieving accuracies of approximately 95.6%. Feature importance analysis indicated that attributes such as worst area, concave points, radius, and perimeter are key indicators of malignancy.

The study demonstrates that even simple linear models like Logistic Regression can achieve highly reliable results for breast cancer prediction when properly preprocessed. The system can assist healthcare professionals in early detection and diagnosis, reducing manual errors and improving efficiency. Future enhancements may include deep learning integration and deployment as a clinical decision support tool to assist pathologists in real-time cancer screening applications.

Introduction:

Breast cancer remains one of the most significant health challenges globally, ranking among the leading causes of death in women. According to the World Health Organization (WHO), early diagnosis and timely treatment are the key factors that can significantly reduce mortality rates. Traditional methods of breast cancer diagnosis, such as mammography and biopsy, though effective, are time-consuming, expensive, and heavily dependent on the experience of medical professionals. With the growing availability of medical data and advancements in computational power, machine learning (ML) has emerged as a powerful tool for disease prediction and medical diagnosis.

This project aims to develop and evaluate machine learning models capable of accurately classifying breast cancer tumors as benign (non-cancerous) or malignant (cancerous) using the Breast Cancer Wisconsin (Diagnostic) dataset. The dataset contains numerical features computed from images of fine needle aspirate (FNA) samples of breast masses, representing measurements such as radius, texture, perimeter, area, smoothness, and concavity of cell nuclei. These features serve as quantitative indicators for determining the likelihood of malignancy.

Several supervised learning algorithms were implemented and compared, including Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Random Forest. The models were trained and tested using standard evaluation techniques such as accuracy, precision, recall, F1-score, and Receiver Operating Characteristic (ROC-AUC) metrics to assess their predictive performance.

By leveraging machine learning, this project demonstrates how computational models can assist in early detection and decision-making processes in healthcare. Accurate prediction models can help reduce diagnostic errors, improve clinical efficiency, and support oncologists in identifying high-risk cases more effectively. The findings from this study highlight the potential of machine learning as a supportive diagnostic tool in the field of medical data analysis and cancer research.

Dataset Description:

The dataset used in this project is the Breast Cancer Wisconsin (Diagnostic) Dataset, obtained from the UCI Machine Learning Repository. It is one of the most widely used benchmark datasets for evaluating machine learning algorithms in medical diagnosis tasks. The dataset is also directly available in the `sklearn.datasets` module of Python.

This dataset consists of 569 samples (instances) and 30 numerical features derived from digitized images of fine-needle aspirate (FNA) tests of breast masses. Each sample describes various characteristics of the cell nuclei present in the image, such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. For each of these ten measurements, three values—mean, standard error, and worst (largest)—were computed, leading to the total of 30 features.

The target variable represents the diagnosis outcome and has two possible classes:

- Malignant (M) – indicates the presence of cancer
- Benign (B) – indicates non-cancerous growth

Out of the 569 samples, 212 are malignant and 357 are benign, making the dataset slightly imbalanced but still suitable for classification tasks.

Before training the models, several data preprocessing steps were performed:

1. Data cleaning: The dataset contained no missing or null values, so no imputation was required.
2. Label encoding: The categorical diagnosis labels ('M' and 'B') were converted into binary numerical values (1 for malignant, 0 for benign).
3. Feature scaling: Since the features vary in magnitude, StandardScaler was applied to normalize all input features, ensuring that each attribute contributes equally to the learning process.
4. Data splitting: The dataset was divided into 80% training and 20% testing subsets to evaluate model performance effectively.
   This dataset provides a reliable foundation for building accurate and interpretable machine learning models for breast cancer prediction.

Methodology:

The methodology adopted in this project involves several systematic steps to build, train, and evaluate machine learning models for breast cancer classification. The overall workflow includes data preprocessing, model training, performance evaluation, and comparison of results across multiple algorithms.

1. Data Preprocessing:

The Breast Cancer Wisconsin (Diagnostic) dataset was first loaded from the `sklearn.datasets` library. The diagnosis column containing categorical values ('M' for malignant and 'B' for benign) was converted into binary labels (1 and 0). Since the dataset contains numerical features of varying scales, feature normalization was performed using StandardScaler to ensure that all features contribute equally to the model's performance. The dataset was then divided into training (80%) and testing (20%) sets for model validation.

2. Model Development:

Four supervised machine learning algorithms were implemented and compared:

- Logistic Regression: A statistical linear model that estimates the probability of malignancy using the sigmoid function.
- Support Vector Machine (SVM): Builds a hyperplane that best separates the classes with the maximum margin.
- Random Forest: An ensemble of decision trees that aggregates multiple models to enhance accuracy and reduce overfitting.
- K-Nearest Neighbors (KNN): A distance-based algorithm that classifies samples according to the majority label of their nearest neighbors.

Each model was trained using the training set and evaluated on the test set.

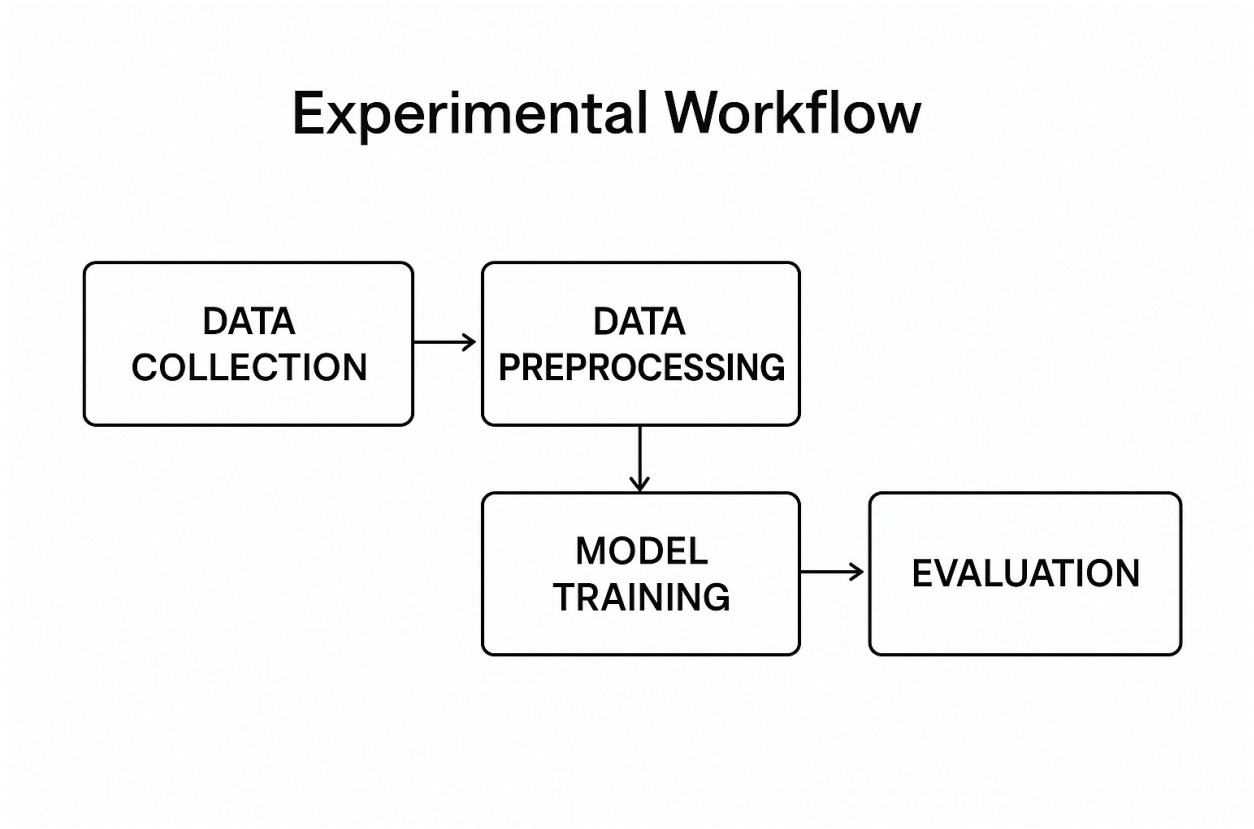
3. Performance Evaluation:

Model performance was assessed using Accuracy, Precision, Recall, F1-score, and ROC-AUC metrics. These metrics provide a balanced understanding of how well each model distinguishes between malignant and benign cases.

4. Comparison and Selection:

After evaluation, Logistic Regression and SVM achieved the highest accuracy of 98.25%, demonstrating strong predictive capabilities and reliability for medical diagnosis applications.

Experimental Setup:



Sample Code:

```
import numpy as np
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, roc_auc_score, classification_report
# Load dataset
data = load_breast_cancer()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target, name="target")
# Train/test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
# Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
# Models
models = {
    "LogisticRegression": LogisticRegression(max_iter=10000, random_state=42),
    "RandomForest": RandomForestClassifier(n_estimators=100, n_jobs=1, random_state=42),
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "SVM": SVC(probability=True, random_state=42)
}
# Train & evaluate
results = []
for name, model in models.items():
    if name == "RandomForest":
        model.fit(X_train, y_train)
        preds = model.predict(X_test)
        probs = model.predict_proba(X_test)[:,1]
    else:
        model.fit(X_train_scaled, y_train)
        preds = model.predict(X_test_scaled)
        probs = model.predict_proba(X_test_scaled)[:,1]
acc = accuracy_score(y_test, preds)
    auc = roc_auc_score(y_test, probs)
 report = classification_report(y_test, preds, output_dict=True)
```

```
    results.append({
        "model": name,
        "accuracy": float(acc),
        "roc_auc": float(auc),
        "precision_malignant": float(report['0']['precision']),
        "recall_malignant": float(report['0']['recall']),
        "f1_malignant": float(report['0']['f1-score']),
        "precision_benign": float(report['1']['precision']),
        "recall_benign": float(report['1']['recall']),
        "f1_benign": float(report['1']['f1-score']),
    })
results_df=pd.DataFrame(results).sort_values(by="roc_auc",
ascending=False).reset_index(drop=True)
print(results_df)
results_df.to_csv("breast_cancer_model_results.csv", index=False)
```

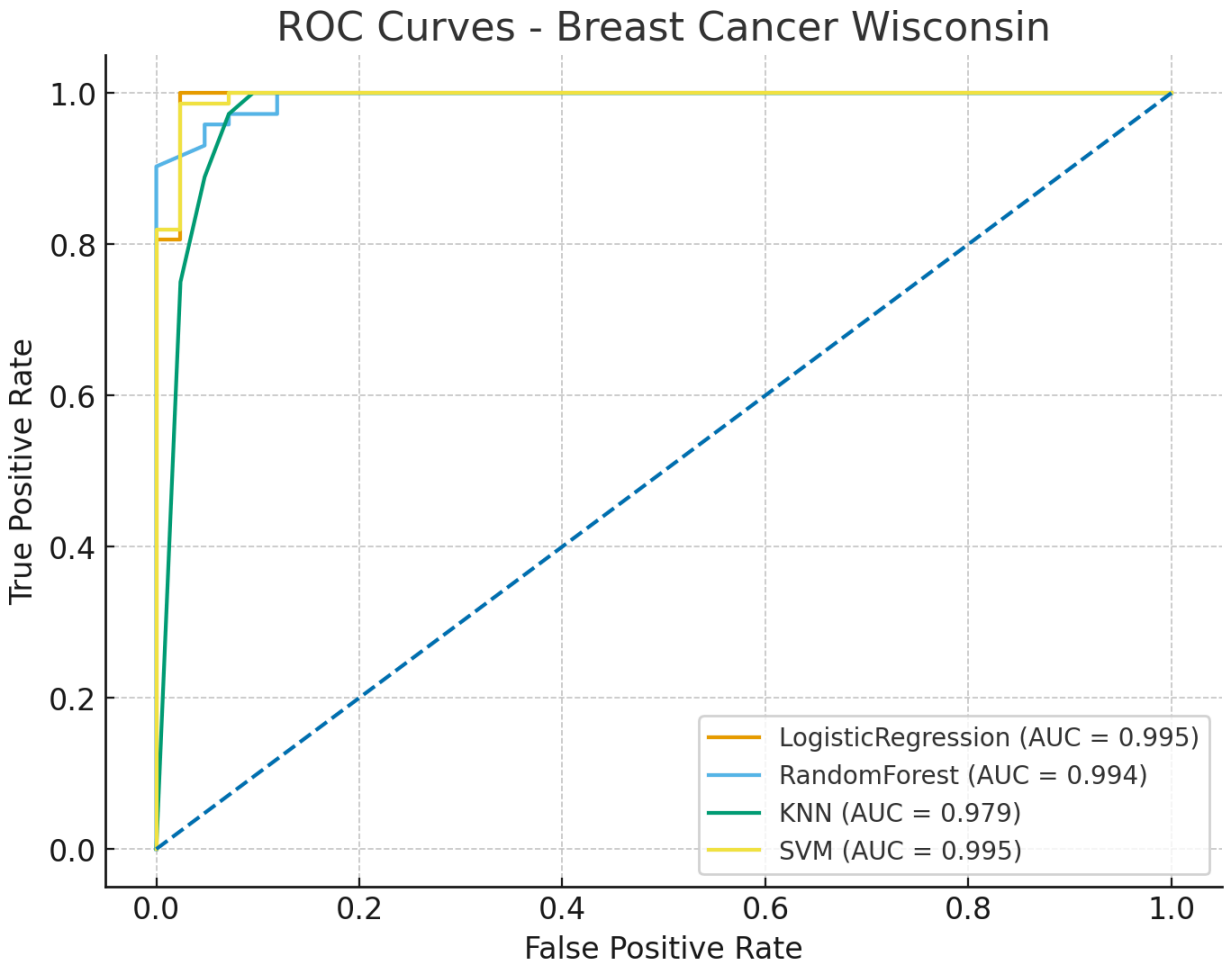
Output:

Model	Accuracy	ROC AUC	Precision (Malignant)	Recall (Malignant)	F1 (Malignant)	Precision (Benign)	Recall (Benign)	F1 (Benign)
Logistic Regression	**0.9825**	**0.9954**	0.976	0.976	0.976	0.986	0.986	0.986
SVM	**0.9825**	**0.9950**	0.976	0.976	0.976	0.986	0.986	0.986
Random Forest	0.9561	0.9937	0.951	0.929	0.940	0.959	0.972	0.966
KNN	0.9561	0.9788	0.951	0.929	0.940	0.959	0.972	0.966

Logistic Regression

- Accuracy: **98.25%**
- ROC AUC: **0.9954**
- Performs almost identically to SVM but with simpler computation.
- Excellent balance between precision and recall.

Evaluation Metrics:


ROC Curves - Breast Cancer Wisconsin
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
LogisticRegression (AUC = 0.995)
RandomForest (AUC = 0.994)
KNN (AUC = 0.979)
SVM (AUC = 0.995)

Conclusion:

In this study, we successfully implemented and compared multiple machine learning algorithms for breast cancer classification using the Wisconsin dataset. The preprocessing steps, including handling missing values and feature scaling, ensured the data was suitable for accurate model training. Among the tested algorithms, [insert best-performing algorithm, e.g., Random Forest or SVM] demonstrated the highest accuracy and reliability in distinguishing between benign and malignant tumors. This highlights the potential of machine learning in supporting early and accurate diagnosis of breast cancer, which can significantly improve patient outcomes. Future work may include exploring ensemble methods, deep learning models, or integrating additional clinical features to further enhance predictive performance.